

GEOGRAPHY

INSTANT LESSONS POWERPOINT

THE ENVIRONMENT & RESOURCE MANAGEMENT

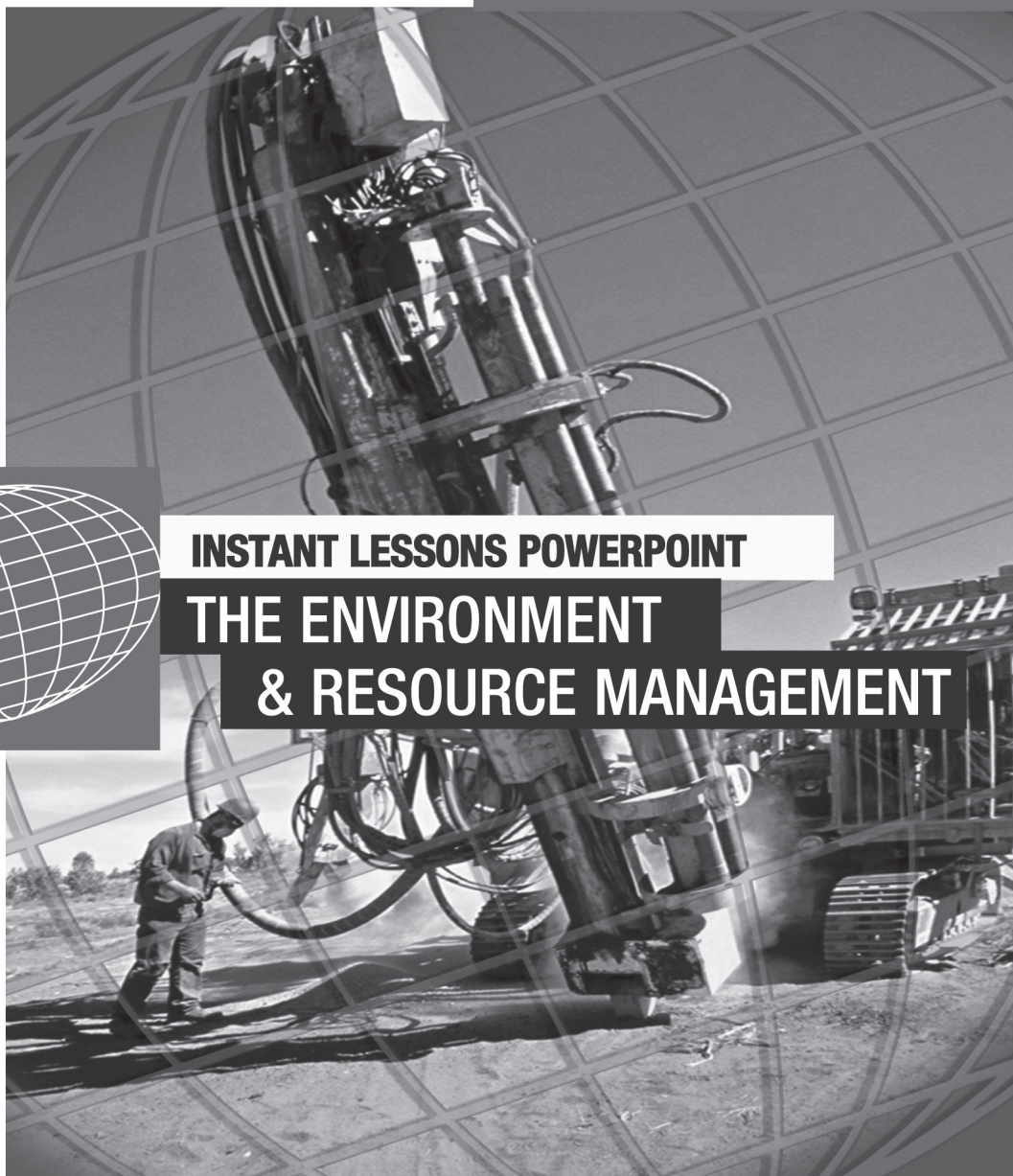


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**THE ENVIRONMENT
& RESOURCE MANAGEMENT**



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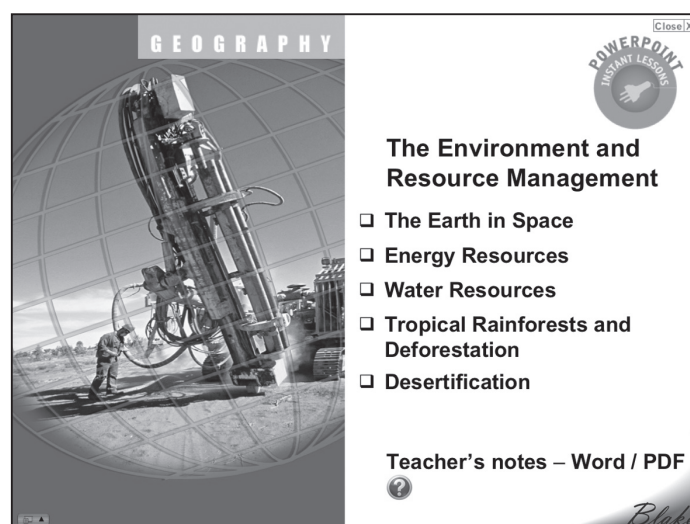
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INTRODUCTION

Instant Lessons PowerPoint slide presentations have been designed to assist teachers in presenting essential geography curriculum to junior and middle secondary school students. Physical, environmental, social and political geography themes are included throughout the series.

The enclosed CD-ROM contains:

- 5 PowerPoint slide presentations
- this book, in PDF format
- the activity sheets in this book, as editable word documents.



Key features and uses:

- Each topic is explored through a core set of slides.
- Teachers can tailor-make their own slide presentations by adding, deleting or altering slides.
- Carefully chosen diagrams, maps, graphs, illustrations and photographs illustrate the concepts being taught.
- On-screen activities test and extend student comprehension.
- Slide notes are provided as a summary of the content of each slide.
- Teacher's notes are provided in this book for each topic.
- Student worksheets have been designed so that they can be used during the presentation or later in the classroom, or as homework, extension, assessment or revision. They can be accessed on the CD-ROM in PDF and Word formats, or photocopied from this book.

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It is recommended that teachers preview the slide sets before showing them to their class, to become familiar with the format and features.

Read the Teacher's notes for an overview of each topic, including key terms and concepts.

The PowerPoint presentations

Starting a slideshow presentation

The Slide Show is used to display the presentation on the screen. It is recommended that you use an external projector for whole-class use.

There are a number of ways to begin a Slide Show presentation. In the Windows environment, the CD-ROM will autorun. From the Main menu, simply click on the presentation you wish to view.

In the Macintosh environment, double-click the file index.pps. From the Main menu, click on the presentation you wish to view.

You can also open each presentation from within the PowerPoint application.

Navigating

Main menu

- ◆ From the Main menu, click Close to quit.
- ◆ During the presentations, click Close in the top right corner at any time to return to the Main menu.

Navigating using the mouse

- ◆ Click – Moves presentation to the next slide.

Moving forward

- ◆ Press the N key.
- ◆ Press the Spacebar or Enter key.
- ◆ Press the Page Down key.
- ◆ Press the Right Arrow key.

Moving backward

- ◆ Press the P key.
- ◆ Press the Backspace key.
- ◆ Press the Page Up key.
- ◆ Press the Left Arrow key.

Navigating using the slide show toolbar

At the bottom left corner of the slide show, a toolbar is displayed. There are three buttons that deal with navigation.

- ◆ Left Arrow button – Moves presentation back to the previous slide.
- ◆ Right Arrow button – Moves the presentation to the next slide
- ◆ Slide Icon button – Provides a pop-up menu with Next and Previous menu items, as well as a Go to Slide item that allows for navigation to a specific slide.
- ◆ Right-Click – Provides a pop-up menu with Next and Previous menu items, as well as a Go to Slide item that allows for navigation to a specific slide.

Navigating to a specific slide

There are a number of different ways to display a specific slide.

- ◆ Slide Number + Enter – While in the Slide Show mode, type in the slide number to be displayed and press Enter.
- ◆ Control-S – To display a menu of all slides in the presentation, press Control and S keys. A menu will appear. Select the desired slide from the menu and press Enter.
- ◆ Slide Icon Button – Choose the Slide Icon button from the Slide Show Toolbar. Choose Go to Slide, then choose the desired slide.
- ◆ Right-Click – Choose Go to Slide, then choose the desired slide.

Blanking out the screen

To present a blank screen while in the Slide Show mode:

- ◆ Slide Icon Button – Choose the Slide Icon button from the Slide Show Toolbar. Choose Screen menu item. Select Black Screen or White Screen.
- ◆ Right-Click – Choose Screen menu item. Select Black Screen or White Screen.

Speaker notes

Speaker notes are included for each slide to provide a summary of the content of that slide. In Normal view, these notes appear at the bottom of each slide.

Speaker notes can be displayed during a slide show presentation. However, when displayed they will be visible to the audience. If you do not want students to see the speaker notes, but need to access them, print the notes before class from the Word document on the CD-ROM.

To view speaker notes during a slide show presentation:

- ◆ Right-click, choose Screen | Speaker Notes to see the notes for this slide.

Pointer / pen

The Pointer and Pen tools can be used in the Slide Show view to highlight or draw on the screen, much as would be done on a whiteboard. Drawings can be saved with the presentation or discarded.

Note that while using Pointer tools during a Slide Show, you will not be able to click the mouse to move to the next slide. Instead, use one of the alternative navigations provided on the previous page, such as the Right Arrow key.

To activate the Pointer tools:

1 In Slide Show view, right-click, choose Screen | Pointer Options (or click on the Pen icon in the Slide Show Toolbar) and choose one of the following.

- ◆ Arrow – provides an arrow on the screen.
- ◆ Ballpoint Pen – used to write notes on the screen.
- ◆ Felt Tip Pen – a wider pointed pen, useful for drawing. (Windows Office 2003 or higher only)
- ◆ Highlighter – used to apply a highlight of color that is transparent. (Windows Office 2003 or higher only)
- ◆ Ink Color – Choose the color to be used with the tool.
- ◆ Eraser – used to erase entries. (Windows Office 2003 or higher only)
- ◆ Erase All Ink on Slide – Removes all entries from the screen. (Windows Office 2003 or higher only)

2 Use the mouse to draw or write on the screen.

Ending a presentation

Each of these actions will end the presentation:

- ◆ Click Close in the top right corner.
- ◆ Press the Esc key.
- ◆ Right-click and choose End Show.
- ◆ Choose the Slide Icon button in the Slide Show Toolbar & select End Show.

Technical specifications Windows

The CD-ROM will autorun in the Windows environment.

To view the presentations – Windows 97 or later operating system

To edit a presentation, open it from within the PowerPoint application.

Note that if editing these PowerPoint presentations in a version prior to Office 2000, cosmetic losses will occur in some animations, and visual effects such as drop shadows may be lost. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Macintosh

Double-click the file index.pps on the CD-ROM. From the Main menu, click on the presentation you wish to view.

You can also open each presentation from within the PowerPoint application.

To view a fully featured presentation, and edit it, you must have the PowerPoint 2004 application installed. (i.e. Office 2004 for Macintosh or later)

To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a Macintosh environment, cosmetic losses will occur in some animations, and visual effects such as drop shadows may be lost. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Teacher's notes on the CD-ROM

To open these files you need:

- ◆ **PDF file** – Adobe Acrobat Reader version 6, or higher.
- ◆ **Word document** – Microsoft Office 2000, or higher.

TOPIC 1 – The Earth in Space

This set of slides uses diagrams, illustration and simple explanations to show how the Earth's tilt impacts on everyday life through the amount of solar radiation places on Earth receive at different times of the day and year.

Key terms: Planetary orbits, the Earth's tilt and its impact on life, latitude and solar radiation

Concepts explored: What is a planet? Earth and other planets in the solar system; orbits; relationships between average distance from the Sun and length of years; diagrams and mapping the solar system; impact of the Earth's tilt on seasons, climate zones and length of days and nights; latitude and solar radiation; Earth's orbit around the Sun; hemispheres

Examples used: Planets in Earth's solar system

Extension slide set: The Earth in Space quiz. This short quiz tests student understanding of the main concepts and definitions in this slide set. Questions require students to apply what they know to short scenarios and to correctly label a diagram with climate zones and the key latitudes.

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TOPIC 2 – Energy Resources

This set of slides examines the main forms of energy used today and how this affects the environment. Students will examine the realities of current patterns of energy use and the manner in which energy will need to be created in the future.

Key terms: Renewable and non-renewable energy sources, patterns of energy use, importing and exporting energy

Concepts explored: Types of energy resources – non-renewable and renewable; why renewable resources continue to be available despite human use; natural replacement; fixed, limited and finite supplies; classifying energy resources; patterns of global energy use; future predictions of energy use; changing patterns of energy use; developed and highly populated countries; calculating energy consumption; the oil crisis, the Greenhouse Effect and pollution; industry sector oil consumption; domestic oil use; world oil reserves; oil – a finite resource; Australian energy imports and exports; calculating Australian energy imports and exports

Examples used: Australian energy statistics

Extension slide sets:

- Set 1 – The Oil Crisis slides examines this finite resource and considers when it may run out and the impact this might have.
- Set 2 – Australian energy imports and exports – a focus on what resources are traded.

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TOPIC 3 – Water Resources

This set of slides introduces the key sources of clean water in the world and the impact a lack of water has on the Australian community. Links between drought and bushfire are made and the causes of both are examined. The management of domestic and rural water use are also illustrated using examples and statistics.

Key terms: Drought and bushfire; water use per capita; the importance of water to society, the economy and the environment

Concepts explored: Sources of water in the world; links between lack of clean water and disease; water use per capita in Australia; percentage use of water in agriculture, urban areas, industry and rural areas; historical water shortages in Australia; drought and bushfire; impact of drought and bushfire on society, the economy and the environment; water management in rural areas; domestic water management

Examples used: Australian examples and statistics

Extension slide set: Monitoring drought in Australia – this series of slides describes what key indicators meteorologists use to declare drought in areas where ‘plentiful’ rain has not occurred. A series of discussion questions are posed to help examine the current water situation in the local region.

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TOPIC 4 – Tropical rainforests and deforestation

This set of slides focuses on key characteristics of rainforests and the importance of them to people and the environment. The causes of deforestation are examined and use of plantation and old growth forest timber is explored through diagrams and statistics.

Key terms: Rainforest, ecosystems, deforestation, old growth forest, plantation timber

Concepts explored: Types of forests; biomes; the nature of tropical rainforests; the importance of rainforests to communities and plants and animals; causes of deforestation; deforestation in Brazil; tropical rainforest climate; Kalimantan forest climate; identifying parts of rainforest ecosystems; causes and impacts of deforestation; arguments for and against plantation and old growth logging; uses of old growth forest timber; wood availability statistics for Australia 2001–2044

Examples used: The Brazilian rainforest, Kalimantan, Indonesia rainforest, Australian logging and wood availability statistics

Extension slide set: Issues to do with plantation and old growth forest cultivation and use, such as the use of wood, availability of wood in the future and changing lifestyles to suit resource availability

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TOPIC 5 – Desertification

This set of slides examines the features of desert environments and the human and natural causes that have resulted in the artificial increase in the amount of desert type areas, world-wide. Ways of combating desertification are also explored.

Key terms: Deserts, erosion, desertification

Concepts explored: Types of deserts; deserts and erosion; what causes erosion in deserts; types of desert landforms – dunes, rock formations, badlands; human and natural causes of desertification; the impacts of desertification on people and the environment; global extent of desertification; populations and land area affected; combating desertification; the United Nations Convention to Combat Desertification

Examples used: World statistics and data that explore the extent of desertification

Extension slide sets: Combating desertification includes the human causes of desertification and the practical measures the UNCCD agreement encourages in order to halt or reverse this process.

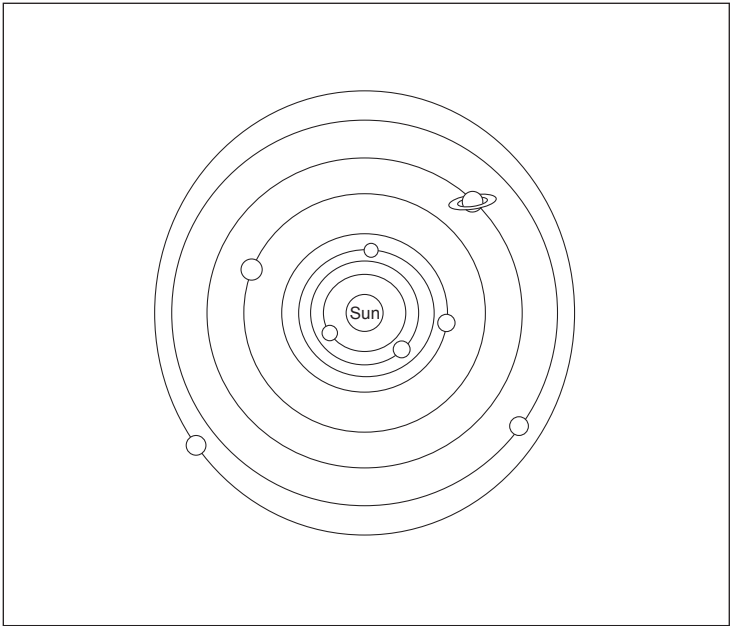
THE EARTH IN SPACE 1

Earth and the solar system

1 Complete the following sentences:
All living and non-living things on Earth are part of a large planet in one of the s_____ s_____ of the Universe. A planet is a l_____, spherical m_____, r_____ in space a_____ the Sun.

2 This diagram shows the orbits of the planets around the Sun. Label each planet: Earth, Jupiter, Mars, Mercury, Neptune, Saturn, Uranus, Venus.

3 What does the table at the right tell you? Use examples to explain.



Planet	Average distance from Sun (million km)	Length of one year (Earth days or years)
Mercury	58	88 days
Venus	108	225 days
Earth	150	365 days
Mars	227	687 days
Jupiter	779	12 years
Saturn	1430	29 years
Uranus	2970	84 years
Neptune	4500	165 years

4 In terms of each planet’s orbit around the Sun, how is a year defined?

5 In terms of the rotation of each planet, how is a day defined?

6 What relationship can you see between the distance from the Sun and the length of one year? Suggest why this is the case.

Distance in the solar system

- 7** Approximately how far is the orbit of Mars from the orbit of Earth?_____
- 8** The orbits of which two planets are closest together?_____
- 9** Between the orbits of which two planets is the largest ‘gap’?_____
- 10** If you were travelling from the Sun to Uranus, approximately what fraction of the total trip would you have done when you reached Saturn’s orbit?

- 11** If you were travelling from the Sun to Neptune, how many planets’ orbits would you have passed when you had completed one-quarter of the trip?

THE EARTH IN SPACE 2

Mapping the solar system

12 Use these words to complete the sentences: *always, different, difficult, Mercury, orbits, tomorrow, year, enormous.*

On Earth, it is easy to make a location map as things stay in the same place compared to each other. If the school crossing is outside the school today, then it will also be there _____, and the next _____.

In the solar system, the planets are all moving around the Sun, but in _____ directions and at different distances from it. The distances between the planets is _____ changing. _____ is always the closest planet to the Sun. But sometimes the closest planet to Earth is Mars and at other times it is Venus.

Instead of mapping where the planets are now, maps of the solar system show the planets' _____ around the Sun. Distances between the orbits are _____, and it is _____ to show scale correctly.

Impact of the Earth's tilt

Earth spins on an axis that passes through the North Pole and the South Pole.

One rotation takes 24 hours. The Earth's axis is inclined at an angle of 23.5 degrees to the Sun. When viewed from above the North Pole, Earth spins in an anti-clockwise direction. The tilt of the Earth is the reason for the different seasons. It is also the reason why climate zones exist on Earth, and why the length of day and night varies. The world's three broad climatic zones are: polar, temperate and tropical.

13 What climate zone does your area fall into? _____

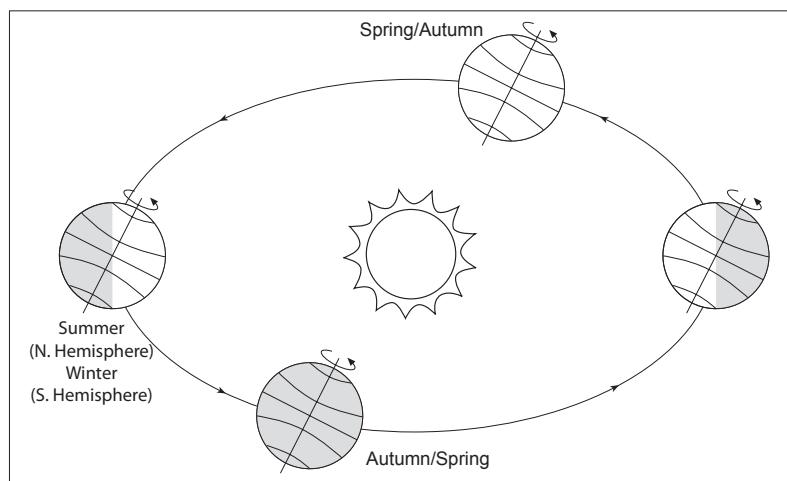
14 What are the characteristics of this climate zone? _____

Earth's orbit of the sun

During one orbit of the Sun, the tilt of the Earth means that different parts of it point more directly at the Sun at different times.

15 When it is winter in the Southern Hemisphere, what is the season in the Northern Hemisphere?

16 When it is autumn in the Southern Hemisphere, what is the season in the Northern Hemisphere?



17 What does the circular arrow indicate? _____

18 Is this diagram drawn to an accurate scale? Why? _____

19 Moscow is a city that receives 16 hours of daylight in August and 8 hours of daylight in January. Which is the warmer month? Why?

20 In which months in Moscow would the hours of daylight be similar? _____

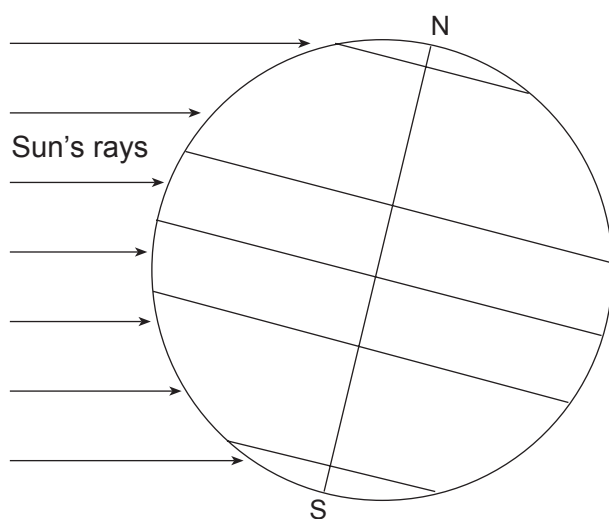
THE EARTH IN SPACE 3

The Earth in space quiz

- 1** The orbit of Uranus is smaller than that of Earth – true or false?_____
- 2** Mercury is a little more than half way between the Sun and Venus – true or false?_____
- 3** Cross out the incorrect word:
As the planets get further from the Sun, the length of one year on each planet gets *longer* / *shorter*.
- 4** The orbit of which planet is closer to Earth's orbit – Mars or Venus?_____
- 5** On Mercury, one year is 88 Earth days long. If you lived there, how many birthdays would you have during the course of one Earth year? (Answer to the nearest whole number.)

- 6** At what angle is the Earth's axis inclined to the Sun?_____
- 7** Complete the sentence: The Earth's tropical climate zone is located between the Tropic of Cancer and the Tropic of _____.
- 8** Cross out the incorrect words:
When it is spring in the northern hemisphere, it is *winter* / *autumn* / *summer* in the southern hemisphere.
- 9** Cross out the incorrect word:
In higher latitudes, the area over which the sun's rays spread is *smaller* / *larger* than the area over which they spread in lower latitudes.
- 10** Compose a question that tests knowledge about seasons in Australia and the tilt of the Sun. Then write the answer to your question.

- 11** Using colour and labels, indicate the locations of the world's three climatic zones: polar, temperate and tropical.



- 12** Correctly label the six lines shown on the Earth diagram above.

ENERGY RESOURCES 1

Types of energy resources

As the world economy continues to grow, the demand for energy is increasing substantially.

- 1 Complete the sentence: Energy resources are classified as r_____ or n_____-r_____.
- 2 What is the difference between these two types of energy resources?

- 3 What is meant by 'finite' and 'fixed' when describing non-renewable resources?

- 4 'Renewable energy resources continue to be available despite human use, due to natural supply, recycling or reproduction.' Use examples to explain what this statement means.

- 5 Why don't we only use energy resources that are perpetually renewable? _____

Patterns of energy use

- 6 From 1980–2004, total world primary energy demand grew by 54% and it is predicted to grow at the same rate until 2030 (OECD, 2006). What factors are driving up energy use?

- 7 What do you think the energy demand growth rates will be in the year 2050 and the year 2500? Why?

- 8 What use do you make of crude oil, coal and natural gas around your home? Around your school?

- 9 What use do you make of renewable energy resources around your home and school?

ENERGY RESOURCES 2

10 Refer to the table below to answer the following questions.

a Calculate the energy use per billion people for each year. (Divide the energy use by the number of billions of people.)

Global population and total energy use	1950	2000
World energy use (energy units per year)	100	429
World population (billions)	2.7	6.0

Source: The World Resources Institute 2004

b Suggest reasons for this increase in energy use. _____

c Calculate the absolute change in energy use and population between 1950 and 2000. (Subtract the 1950 figure from the 2000 figure.)

d Calculate the percentage change in energy consumption between 1950 and 2000. Repeat for the population change. (Divide the absolute change for each by the 1950 figure, and multiply by 100.)

e What do these figures (your answers for questions a–d) tell you? _____

The oil crisis

The world’s dependence on crude oil as a major source of energy grew rapidly during the last century. Economic development, industrialisation and the technology of the internal combustion engine have provided a growing market for crude oil. It is expected that oil prices will continue to rise this century as growing demand exceeds dwindling supplies. Most serious of all is the environmental damage caused by the use of oil. The use of petrol, diesel and other forms of oil contributes to greenhouse gas build-up in the atmosphere. Apart from its impact on global warming, the use of oil also releases toxic pollutants into the atmosphere. These include carbon monoxide, oxides of nitrogen and volatile organic compounds.

11 What three factors have driven up the market for crude oil? Provide examples in your answer.

12 Describe how the way we use oil damages the environment. _____

13 Explain briefly how the use of oil contributes to greenhouse gases. _____

14 What effect can carbon monoxide have on humans? _____

15 What do the words ‘volatile’ and ‘organic’ mean as they are used above? _____

16 How do you think Australia will be affected by running out of oil? _____

ENERGY RESOURCES 3

Australia – an exporter of energy

Australia exports energy resources. China and Japan are large markets for Australian energy.

Major Australian exports, 2005–6	(A\$m)
Coal	23 272
Iron ore	14 381
Non-monetary gold	9 143
Crude petroleum	6 677
Aluminium ores	6 071

Source: Dept of Foreign Affairs and Trade

- 1

What is crude petroleum?
- 2

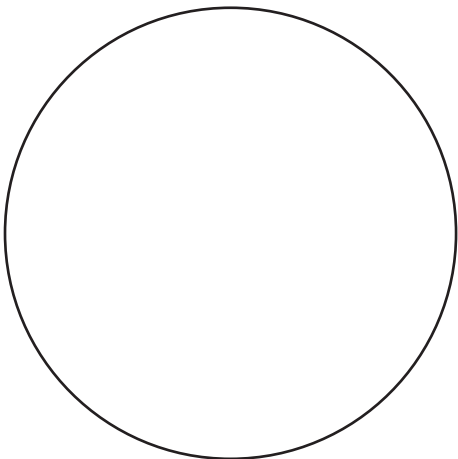
Which of the listed exports are energy resources?
- 3

Graph the 2005–6 Australian exports as a pie chart in the circle at the right. Write a title for your graph.
- 4

What percentage of income from major exports came from energy resources in 2005–06?
(Divide the \$ figure for coal and petroleum by the total \$ figure, \$59 544 million, and multiply by 100.)
- 5

Investigate what the most recent export figures are and their dollar worth. In your workbook, create a pie chart of your findings.
- 6

Compare and contrast the 2005–6 figures with more recent figures. How has energy exportation changed? Search ‘Australian exports’ at www.australia.gov.au/268 and www.abs.gov.au/.



Australia – an importer of energy

Australia also imports energy resources.

Major Australian imports, 2005–6	(A\$m)
Crude petroleum	13 289
Passenger motor vehicles	12 573
Refined petroleum	8 692
Computers	6 413
Telecommunications equipment	6 163

Source: Dept of Foreign Affairs and Trade

- 7

Which of these imports are energy resources?
- 8

What percentage of expenditure on major imports was on energy resources in 2005–06? (Divide the \$ figure for petroleum by the total \$ figure, \$47 130 m, and multiply by 100.)
- 9

Use the websites suggested in question 6 to find out the top five types of materials Australia is presently importing. Compare and contrast the figures for energy resources and investigate where these resources are coming from.
- 10

What is the present dollar value of energy imports in Australia?
- 11

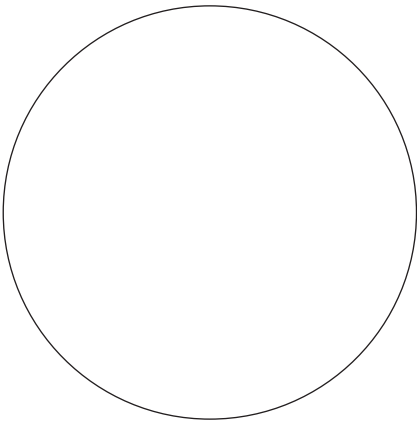
Energy resources discussion: How do you think the global energy crisis could be resolved? Consider how your ideas will affect the environment, economies and business, and people’s lifestyles – including your own!

WATER RESOURCES 1

Global water resources

- 1 Complete the following sentences:
- a The majority of water on Earth is found frozen in _____ and it is _____ water.
 - b Thirty percent of Earth's water is found in _____ or _____.
 - c Less than _____ of the world's fresh water is accessible for _____.
 - d One _____ of the world's population does not have access to safe water.
 - e About _____ deaths occur each day in the world due to _____.

- 2 Using the following data, create a pie chart titled 'Water use in Australia' in the circle at the right.
- Irrigated agriculture (about 75%)
 - Urban and industrial purposes (about 20%)
 - Rural areas – stock and domestic purposes (5%)

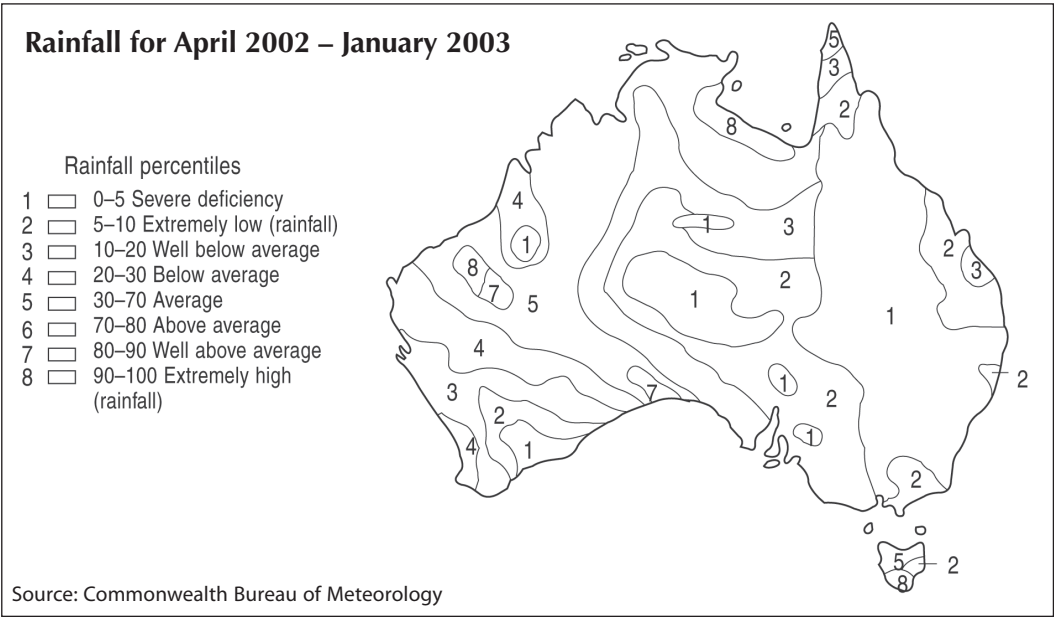


- 3 Australians are some of the largest water consumers in the world. Suggest reasons for this.
- _____
- _____

The importance of water and water shortages

This map shows the pattern of rainfall in Australia for a given period.

- 4 Complete the choropleth map by colouring the zones. Use darker to lighter warm colours for 1–4, blue for 8, greens for 7–6, and leave 5 blank.



- 5 In which regions was the amount of rainfall well below average or worse?
- _____
- 6 Estimate the percentage of Australia that had below average (or worse) rainfall. _____
- 7 What about your region – what was its rainfall like in this period? _____
- _____

WATER RESOURCES 2

8 Compare the rainfall in your region in 2002–03 to the present day. _____

9 Investigate the rainfall totals for the past 12 months in Australia. Colour and label this map to show areas of very high and very low rainfall. Include a title and a key.

Water shortages – drought and bushfire



10 Cross out the incorrect words to make these sentences true.

- a Droughts can *increase* / *decrease* the chances of bushfires.
- b Our natural vegetation is vulnerable to fire, as it is highly *flammable* / *eroded*.
- c Bushfires affect *all states* / *only coastal states* of Australia.
- d Bushfires tend to be *most* / *least* common in the heavily vegetated areas in the *south* / *west*, which are often very dry for long periods over *spring* / *summer*.

11 Complete this sentence: The impact of bushfires is measured in terms of the effect they have on p_____, the e_____ and the e_____.

12 True or false? Australia is the driest inhabited continent, therefore droughts are regular occurrences in various parts of the country._____

13 True or false? Because much of Australia’s native vegetation is adapted to dry conditions, in hot dry periods it is vulnerable to bushfire._____

14 True or false? Scientists believe that global warming, combined with naturally changing weather patterns, will cause longer and more severe droughts in certain parts of Australia._____

Impacts of drought and bushfire

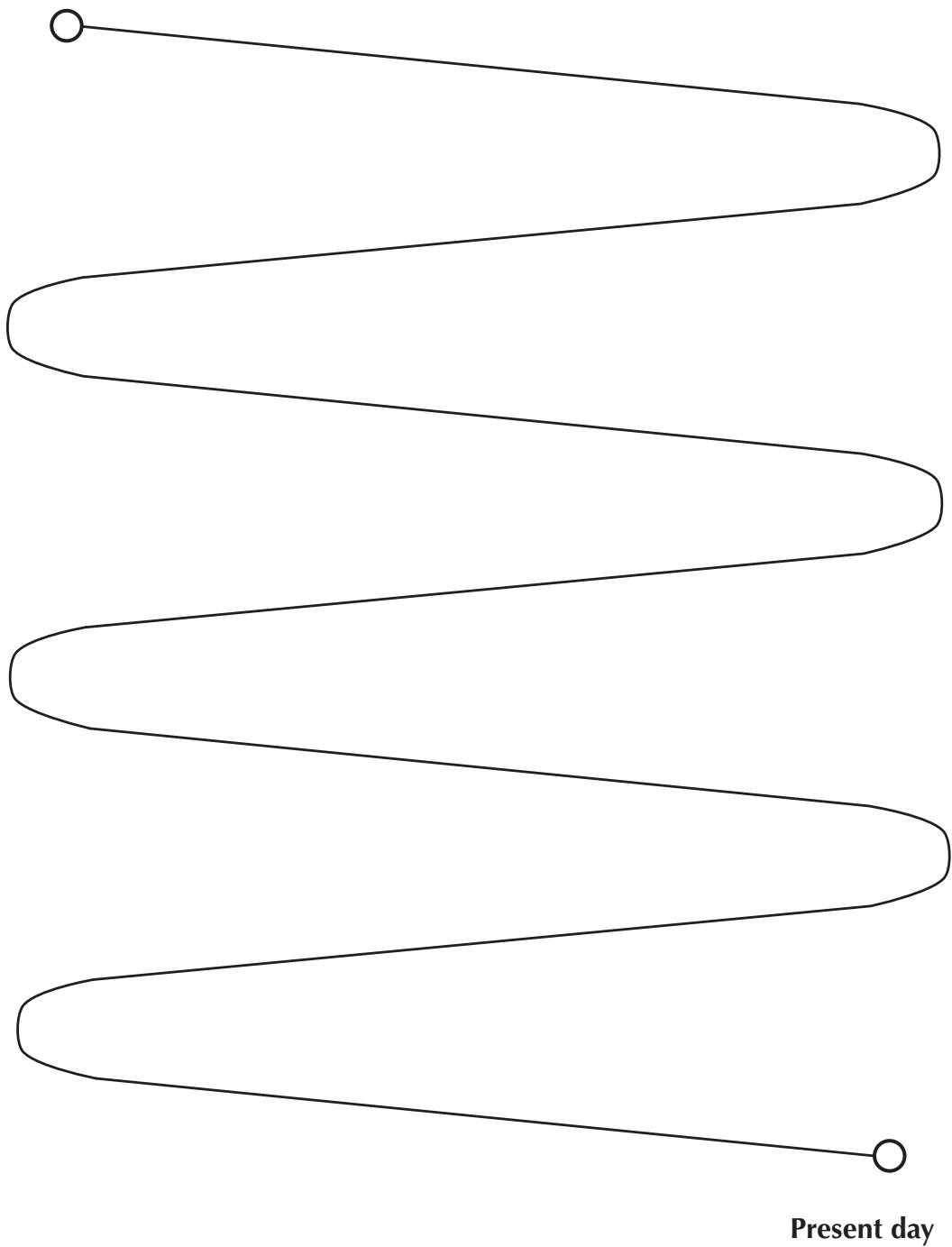
15 Complete the table with examples to show how drought and bushfire impact on Australia. ‘Society’ may mean local communities, or in a wider sense, states or Australia as a whole.

Impact area	Society	Economy	Environment
Crops and crop production			
Livestock farming			
Farm produce prices and farm incomes			
Rural town economies			
Government policies			
Major banks and insurance companies			
Urban water supply			
Native vegetation and fauna			

Drought in Australia

Meteorologists monitor and measure drought severity in terms of rainfall deficiencies for different regions. Research has shown that severe drought affects some part of Australia on average once in 18 years. However, periods between severe droughts have varied between four and 38 years. Because of the variation in climate across the country, there is little chance that the entire country would all be drought-affected at the same time.

Use the Internet to investigate six major drought occurrences in Australian history during white settlement. Mark the date on the timeline below and describe in dot points for each example the location, length and impact of the drought. Search phrases such as 'drought in Australia' and 'drought statistics' as well as the Australian Bureau of Statistics website (www.abs.gov.au).



TROPICAL RAINFORESTS AND DEFORESTATION 1

Forest resources

Forests cover about 30% of Earth's land area. There are three broad forest biomes – tropical, temperate and boreal.

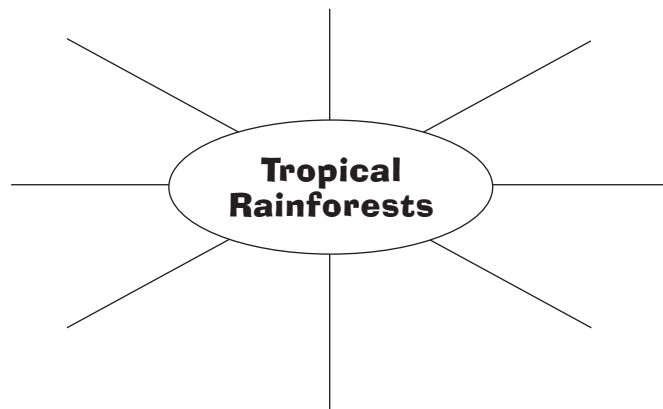
- 1 Complete this sentence: A biome is one of the world's major natural communities, classified according to the _____, and the _____ and _____ life that has adapted to it.
- 2 For each type of forest, write a short definition and provide a location example.
 - a Tropical _____
 - b Temperate _____
 - c Boreal _____

Tropical rainforests

Tropical rainforests are found in the warm, wet climates of the tropics. They are vitally important, not only for their biodiversity, but for the role they play in regulating the global climate.

- 3 How do you think rainforests affect the global climate? _____

- 4 Write your own ideas on this diagram to show the importance of tropical rainforests, locally and globally.



- 5 Complete the following sentences using these words: *infrastructure, leaching, ecosystems, biodiversity, removal, ecological*.

Tropical rainforests are home to a wide range of plants and animals, and play an important part in maintaining global _____. They have many different _____ where communities of plants and animals exist. There are increasing demands on the use of rainforest resources, and widespread _____ of trees is occurring at an alarming rate. Deforestation also occurs with the development of _____ in rainforests, such as roads, railway, water and power supplies. When trees are removed, it promotes _____ of nutrients from the topsoil, which decreases fertility and can inhibit future plant growth. Because so many plants and animals rely on tropical rainforests for their survival, deforestation changes the whole _____ balance.

TROPICAL RAINFORESTS AND DEFORESTATION 2

Deforestation in Brazil

Brazil is the country with the greatest area of untouched tropical rainforest in the world. Between 2000 and 2006, Brazil lost nearly 150 000 km² of rainforest. This is more than twice the area of Tasmania. In many tropical countries, most deforestation is due to the actions of poor, subsistence farmers. But in Brazil, only one-third of recent deforestation can be attributed to this.

6 a Complete this table with examples of each reason for deforestation in Brazil. You may need to research this topic. Search using words such as ‘Amazon’, ‘Brazil’, ‘Deforestation’.

Other reasons for deforestation in Brazil include:	Examples
Land clearing for pasture by commercial interests	
Poor government policies that offer limited protection for the forests	
Inappropriate World Bank projects	
Commercial exploitation of forest resources	

b List two more reasons for deforestation in Brazil in the table above, and include examples.

The rainforest ecosystem

In a rainforest ecosystem there is a complex relationship between plants and animals. They are all interdependent. Also vital for the survival of rainforests are the non-living things – such as sunlight, rain, oxygen, carbon dioxide and decomposing organic matter.

- 7** Write each part of the rainforest ecosystem beside the correct description: *carnivores, epiphytes, herbivores, leaf litter, lianas, parasites, saprophytes, humus.*
- a** Leaves from trees and plants that have fallen to the forest floor and are decomposing to form forest soil that is rich in minerals and nutrients
- _____
- b** Plant-eating animals_____
- c** Meat-eating animals_____
- d** Plants that climb upon a host tree and live off their sap_____
- e** The rich top layer of soil, made of decomposed plant or animal matter_____
- f** Rope-like vines that twist themselves around rainforest trees in search of sunlight
- _____
- g** Fungi such as mushrooms and toadstools that live on dead organic matter found on the forest floor
- _____
- h** Plants such as ferns and orchids that grow on other plants for support_____

TROPICAL RAINFORESTS AND DEFORESTATION 3

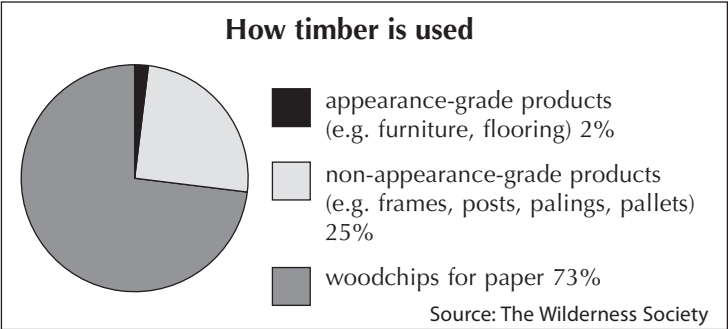
Plantation versus old growth

Old-growth forests contain trees that have grown naturally in a specific location for hundreds of years. Plantation forests contain trees that have been cultivated. Depending on the type of trees, they grow for a few years to a few decades before they are harvested.

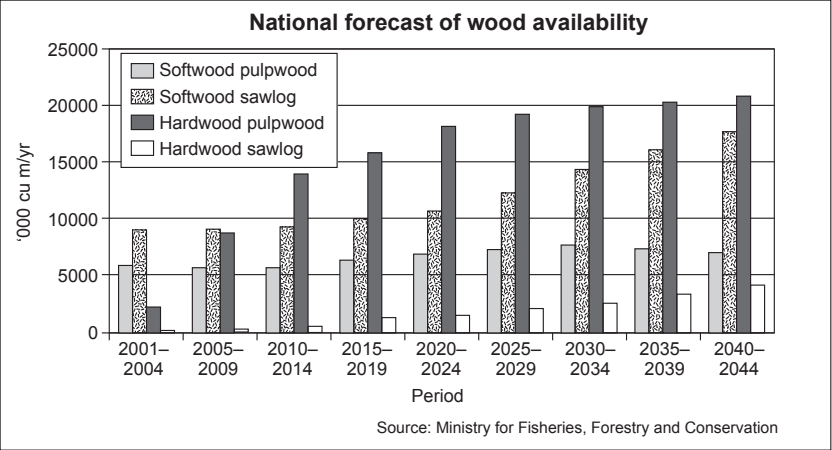
1 List arguments for and against the logging of old-growth forests and the cultivation of plantation forests.

Logging old growth forests		Cultivation of plantation forests	
For	Against	For	Against

2 Comment on what this pie chart tells you about the use of old-growth forest wood in Australia and give your opinion of this usage.



3 In 2002, the Australian government released a report called ‘Plantations of Australia Wood Availability 2001–2044’. This graph is based on data from that report. What does it tell you?



Note that pulpwood is suitable for woodchips and sawlogs are suitable for most other products, except appearance-grade.

4 What does this graph suggest about the way our timber resources are managed?

5 Suggest why forecasts about timber resources are made over decades, rather than years.

DESERTIFICATION 1

Introduction to deserts

Across our globe there are many dry areas that receive 250 mm or less total annual precipitation.

1 Deserts can be hot or cold.

a List three examples of hot deserts.

b List three examples of cold deserts.

Deserts and erosion

2 Complete these sentences:

In hot deserts, the w_____ and r_____ form the land into different shapes.

Although there is not much water, when it falls it is usually in heavy s_____.

The rain strikes the h_____ ground. It does not have time to s_____ in, so runs off in wild torrents that cause a lot of e_____.

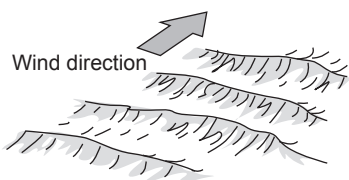
3 Define *erosion* in your own words. Provide some examples from your local area. What has caused the erosion in your area?

Deserts, erosion and landforms

The wind in deserts is usually strong. It contains millions of sand or rock particles, which also cause erosion. Eroded material is then deposited. Desert landforms are the result of this erosion and deposition. Deposition occurs when eroded material is carried away and then settles in a new place.

4 Write the correct landform (listed and defined on the next page) below each diagram.

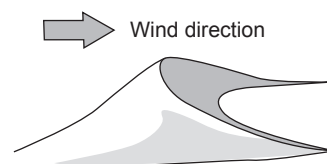
a



b



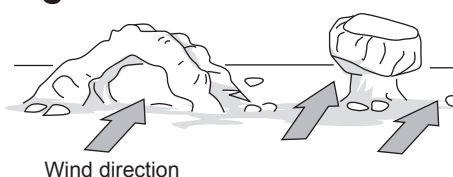
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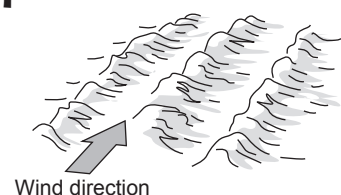
d



e



f



DESERTIFICATION 2

- longitudinal dune – A sand dune that forms as a ridge, parallel to the wind direction
- barchan dune – Formed when sand blasts up the windward side, then falls down the leeward side. The edge is bent by the wind
- arch and mushroom rock – Formed by sand blasting, where the wind throws material against rocks which gradually wear away into interesting shapes
- star dune – Formed when the winds blow from all directions with almost equal force
- transverse dune – A sand dune that forms as a ridge, at right angles to the wind direction
- badlands – Formed when very hilly country is heavily eroded by severe storms and flash floods

5 Research each landform type and list a world location example above each diagram on the previous page.

Impacts of desertification

Desertification is the degradation of land in arid, semi-arid and sub-humid areas. It results from various natural and/or human processes, which make it arid and less productive.

- 6 What natural processes could contribute to desertification? _____
- 7 What human processes could contribute to desertification? _____

Global extent of desertification

This table shows the land and people affected by desertification in each continent.

- 8 Complete the table by calculating:
- a the total area affected by desertification.
 - b each continent's percentage of total area affected by desertification.
 - c the total number of people affected by desertification.
 - d each continent's percentage of the total population affected by desertification.
- 9 Rank the continents in order from most affected by desertification to least affected.

Areas and populations affected by desertification

Region	Area ('000 km²)	%	Population (billions)	%
Africa	7409		108.00	
Asia	7480		123.00	
Australia	1123		0.23	
Europe	296		16.50	
North America	2080		4.50	
South America	1620		29.00	
WORLD				

Source: UNEP

- 10 Explain how you decided on your rankings.

DESERTIFICATION 3

Combating desertification

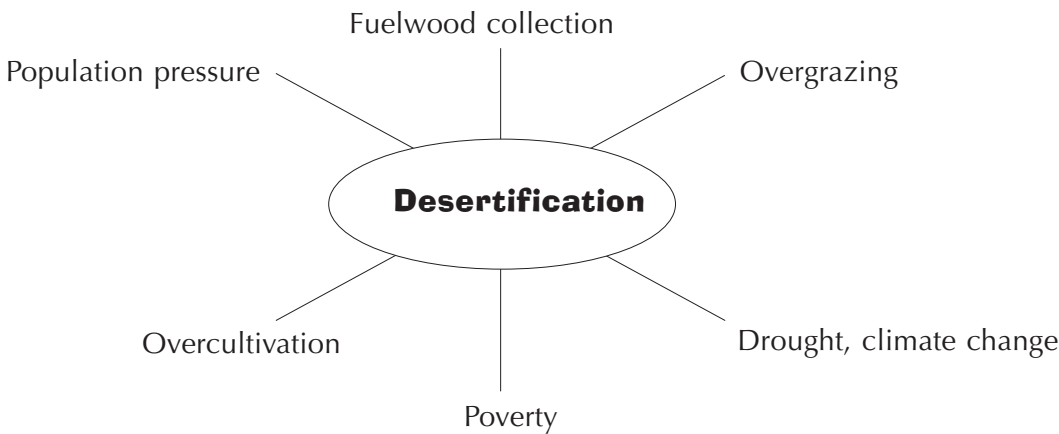
The United Nations Convention to Combat Desertification (UNCCD) has around 200 signatory countries. It administers funding, research, implementation and evaluation for the fight against desertification. Different measures are in place to fight desertification.

1 Explain how each of these measures may help reduce desertification.

Measures to fight desertification	How this helps
Using remote sensing (satellite imagery)	
Using computer simulation models	
Planting rows of trees or shrubs	
Bringing in hay for stock to consume	
Providing local people with simple solar ovens	
Converting land from grazing to cropping	
Placing stones around the bases of trees	
Developing non-agricultural industries	
Limiting or banning the use of 4WDs in fragile areas	

Causes of desertification

2 Describe how each of these factors contributes to desertification.



SELECTED ANSWERS

The Earth in Space

Worksheets 1–2

- 1** (In order) solar systems, large, mass, revolving, around
- 2** (In order from the sun) Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune
- 3** This table shows the correlation between a planet's average distance from the Sun and the length of that planet's year, relative to one Earth year. For example, Mercury is 58 million km from the Sun and its year is 88 Earth days long. Neptune is 4500 million km from the Sun and its year is 165 Earth years long
- 4** The time (in Earth days or years) it takes the planet to orbit once around the Sun
- 5** The time it takes the planet to rotate once on its own axis
- 6** The greater the distance from the sun, the longer the year
- 7** 77 million km
- 8** Venus and Earth
- 9** Uranus and Neptune
- 10** A bit less than half the trip
- 11** 5
- 12** (In order) tomorrow, year, different, always, Mercury, orbits, enormous, difficult
- 13** Teacher to check
- 14** Teacher to check
- 15** Summer
- 16** Spring
- 17** The direction of Earth's rotation
- 18** No, the Sun is many times larger than the Earth, and the distance covered by the orbit is so great, especially compared to the diameter of a planet, that it is difficult to show to scale
- 19** August, because the daylight hours are longer
- 20** March and September

Worksheet 3

- 1** False
- 2** True
- 3** Correct word is longer
- 4** Venus is closer

- 5** 4
- 6** 23.5 degrees
- 7** Capricorn
- 8** Correct word is Autumn
- 9** Correct word is larger
- 10** Teacher to check
- 11** Polar is represented by the two outer areas, near the north and south poles. Temperate is represented by the next two sections further in from Polar. Tropical is represented by the two middle areas
- 12** (In order from North to South) Arctic circle, Tropic of Cancer, Equator, Tropic of Capricorn, Antarctic circle. The sixth line (from North to South) is the Earth's axis

Energy Resources

Worksheets 1–2

- 1** Renewable, non-renewable
- 2** Renewable energy comes from replenishable resources. Non-renewable energy is derived from a source that only consists of a finite amount of product
- 3** Finite describes the state of the limited sources that non-renewable energy is derived from. Fixed explains that there is only a certain amount of a particular product used to create non-renewable energy
- 4** This statement explains that renewable energy is to some extent self-replenishing. A good example of this would be solar energy, or firewood from renewable sources
- 5** Teacher to check. Students should consider issues of cost, availability, efficiency and technological development
- 6** Factors driving up energy use figures include: the industrial revolution in the 18th and 19th centuries, growth in the use of power in the home, increase in the usage of automobiles and aeroplanes

- 7** Teacher to check
- 8** Teacher to check
- 9** Teacher to check
- 10a** 1950: 37.04 energy units per billion people, 2000: 71.5 energy units per billion people
- b** Teacher to check
- c** World energy use = 329 energy units per year, World population = 3.3 billion
- d** Percentage change in world energy use = 329 energy units/year, Absolute change in population = 122%
- e** Teacher to check
- 11** Economic development, industrialisation, the technological discovery of the internal combustion engine
- 12** Contribution to greenhouse gases in the atmosphere
- 13** By releasing toxic pollutants into the atmosphere
- 14** Symptoms include headaches and flu-like effects; larger exposures can lead to significant toxicity of the central nervous system and heart. Following poisoning, long-term serious kidney damage and even death may occur
- 15** Volatile: Highly unstable
Organic: From a naturally traced source
- 16** Teacher to check

Worksheet 3

- 1** Oil
- 2** Coal and crude petroleum
- 3** Teacher to check
- 4** 50.3%
- 5** Teacher to check
- 6** Teacher to check
- 7** Crude petroleum, refined petroleum
- 8** 46.6%
- 9** Teacher to check
- 10** Teacher to check

SELECTED ANSWERS (cont'd)

Water Resources

Worksheets 1–2

- 1a** Polar icecaps, salt
b Soil, underground
c 1%, human use
d Sixth
e 5000, poor sanitation
2 Teacher to check
3 Teacher to check
4 Teacher to check
5 In central Australia and along the East coast
6 Roughly 80%
7 Teacher to check
8 Teacher to check
9 Teacher to check
10a Correct word: increase
b Correct word: flammable
c Correct words: all states
d Correct words: most, south, summer
11 People, environment, economy
12 True
13 True
14 True
15 Teacher to check

Tropical Rainforests and Deforestation

Worksheets 1–2

- 1** Vegetation, plant, animal
2a Tropical rainforest: A hot, humid environment within the Tropic of Cancer and the Tropic of Capricorn. Tropics have a wet season, a cool dry season and a warm dry season
b Temperate: Temperate latitudes of the globe lie between the tropics and the polar circles. The changes in these regions between summer and winter are generally subtle: warm or cool, rather than extreme hot or cold. However, a temperate climate can have very unpredictable weather
c Boreal: A set of forest ecosystems that survive in the polar regions (Antarctic and Arctic regions)

- 3** Teacher to check. Could include improvement of the ozone layer, biodiversity and the environment
4 Teacher to check
5 (In order) biodiversity, ecosystems, removal, infrastructure, leaching, ecological
6a Teacher to check
b Teacher to check
7a Leaf litter
b Herbivores
c Carnivores
d Parasites
e Humus
f Lianas
g Saprophytes
h Epiphytes

Worksheet 3

- 1** Teacher to check
2 Teacher to check
3 This chart gives us the national forecast of wood availability between 2001 and 2044. It shows a large growth in the availability of hardwood pulpwood, and a steady growth in other wood types
4 Teacher to check
5 Because of the time it takes different trees to grow to maturity

Desertification

Worksheets 1–2

- 1a** Teacher to check. Examples might include Sahara, Great Sandy Desert, Kalahari Desert
b Teacher to check. Examples might include Antarctica, Halendi, Ordos
2 (In order) wind, rain, storms, hard, soak, erosion
3 Teacher to check
4a Transverse dune
b Badlands
c Barchan dune
d Star dune
e Arch and mushroom rock
f Longitudinal dune
5 Teacher to check

- 6** Lack of rainfall due to weather patterns, normal climate for an area, area shielded from rain by mountainous terrain nearby
7 Overpopulation, overgrazing of herding animals, use by nomadic peoples, climate change caused by industrialisation
8a 20 008 km²
b Africa 37.03%, Asia 37.38%, Australia 5.61%, Europe 1.48%, North America 10.40%, South America 8.10%
c 281.23 billion
d Africa 38.40%, Asia 43.74%, Australia 0.08%, Europe 5.87%, North America 1.60%, South America 10.31%
9 Teacher to check
10 Teacher to check

Worksheet 3

- 1** (In table order)
 - Helps early identification of dryland areas that are under threat of desertification
 - Helps to predict future events, therefore assisting with measures to prevent desertification
 - These serve as windbreaks, which protect fertile areas from harsh, sandy winds
 - Reduces the need for grazing
 - Reduces the need for fuelwood collection, so trees stay alive and erosion is reduced
 - Crops can help reduce the erosion of soil. Grazing removes the vegetation and increases erosion
 - Reduces erosion around the tree, helps to keep it alive
 - Can provide a more reliable source of income for people, helping to reduce poverty
 - Reduces damage to the environment that 4WDs can cause by destroying vegetation, compacting soil and increasing erosion**2** Teacher to check

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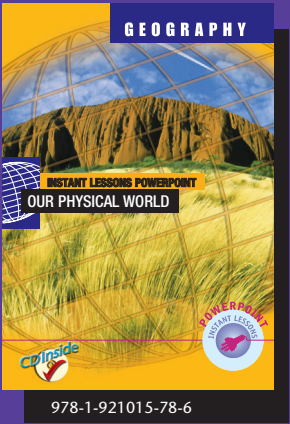
The Environment and Resource Management

This title of five slide sets and the accompanying notes and worksheets explore topics to do with Earth's unique characteristics, its place in the solar system and specific Australian and world-wide resource management issues.

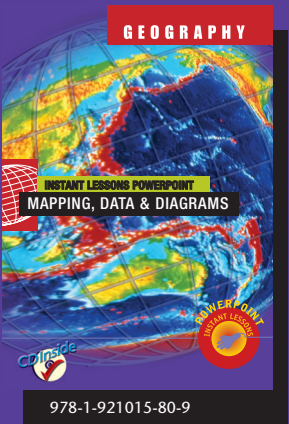
The characteristics of renewable and non-renewable energy resources, patterns of energy use and environmental impacts, as well as Australia's role as an exporter of energy are explained. Australia's propensity for fire and drought are linked to rainfall monitoring and prediction techniques, and how these hazards impact on society, the economy and the environment. Different types of rainforests and their characteristics, as well as how they are exploited, are also illustrated. The final slide set looks at what causes erosion and how this contributes to desertification. Strategies to combat and reverse the impact are explored, as well as the effects on communities around the world.

Slide sets	
1: The Earth in Space 2: Energy Resources 3: Water Resources	4: Tropical Rainforests and Deforestation 5: Desertification

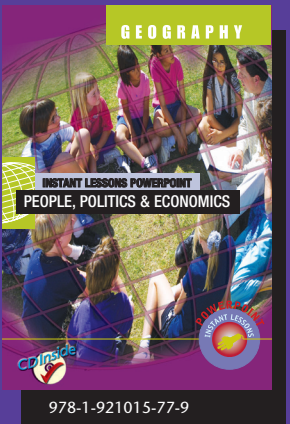
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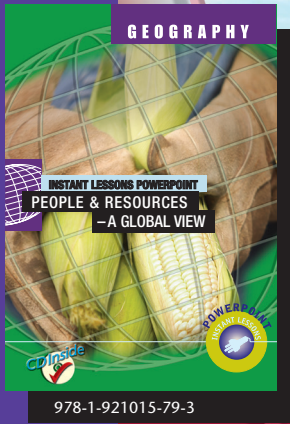
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